IoT Integrated Accessible Smart Home System

Hands-free smart home interaction using eye tracking and IoT

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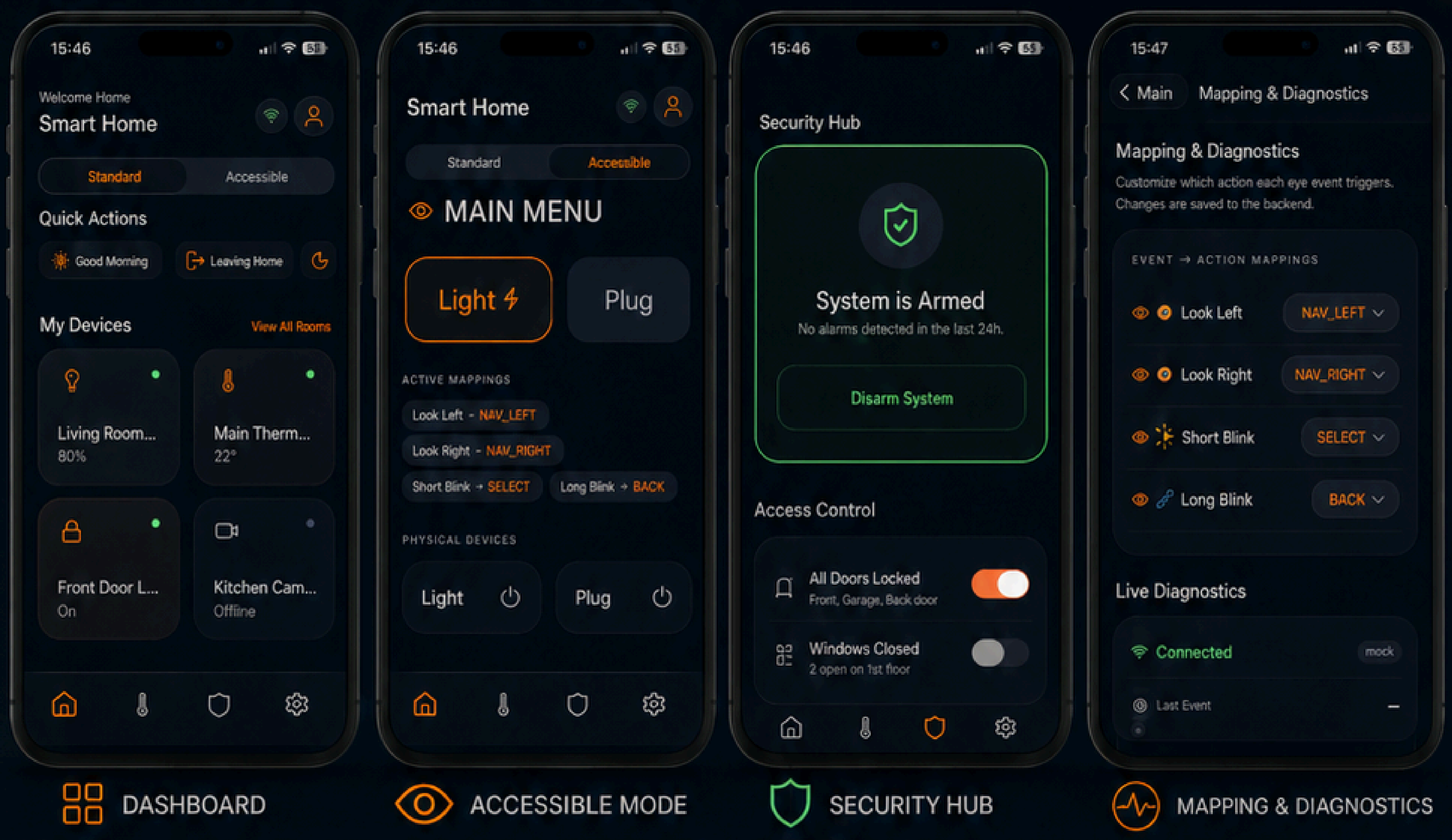


DASHBOARD · ACCESSIBLE MODE · SECURITY HUB · MAPPING & DIAGNOSTICS

1 Project Overview

An accessible smart home system that enables users to control devices through gaze direction and blink gestures. Designed especially for individuals with limited mobility while remaining useful for general users.

2 Core Features

- Eye tracking and blink-based commands
- Camera-based hand gesture control
- Standard and accessible control modes
- Manual fallback control
- Real-time local processing
- Smart device integration over LAN

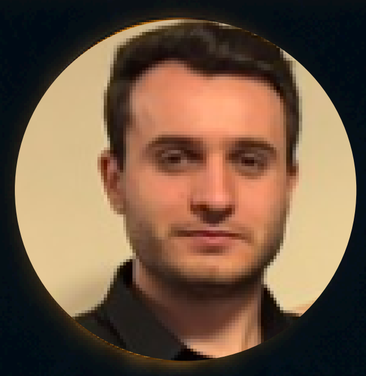
3 System Flow

IP Camera / Smartphone Stream → Raspberry Pi 5 → Eye & Gesture Detection → Intent Mapping → ESP32 / Smart Devices

4 Technologies

- Raspberry Pi 5
- ESP32
- Python
- OpenCV
- MediaPipe Face Mesh
- Wi-Fi / LAN

Team & Advisor


Ömer Altintaş

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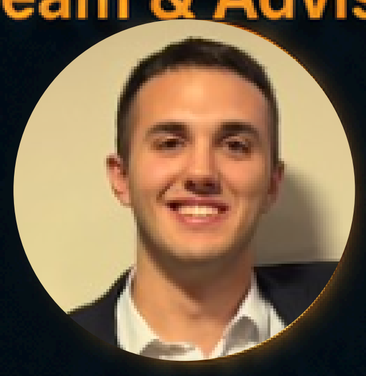
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